

Handouts-3_Contour Integration

Contour Integration

* Evaluate the following by using the method of contour integration.

$$\int_0^{2\pi} \frac{d\theta}{2+\cos\theta}$$

Solⁿ: Let us consider the unit circle $|z|=1$ as the contour C . Then $z=e^{i\theta} \Rightarrow dz=i e^{i\theta} d\theta = iz d\theta$

$$\therefore d\theta = \frac{dz}{iz}; 0 \leq \theta \leq 2\pi.$$

$$\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$$

$$= \frac{1}{2}\left(z + \frac{1}{z}\right)$$

$$= \frac{z^2+1}{2z}$$

$$\begin{aligned}\therefore \int_0^{2\pi} \frac{d\theta}{2+\cos\theta} &= \oint_C \frac{\frac{dz}{iz}}{2 + \frac{z^2+1}{2z}} \\ &= \frac{1}{i} \oint_C \frac{\frac{2z}{2}}{4z + z^2 + 1} dz \\ &= \frac{2}{i} \oint_C \frac{dz}{z^2 + 4z + 1} \\ &= \frac{2}{i} \oint_C f(z) dz, (\text{say}) \quad \text{--- (1)}\end{aligned}$$

$$\text{where } f(z) = \frac{1}{z^2 + 4z + 1}$$

The poles of $f(z)$ are obtained by solving the eqn

$$z^2 + 4z + 1 = 0$$

$$\Rightarrow z = \frac{-4 \pm \sqrt{16-4}}{2}$$

$$= -2 \pm \sqrt{3}$$

$$|z| = |-2 + \sqrt{3}| = 0.27 < 1$$

$$\text{and } |z| = |-2 - \sqrt{3}| = 3.73 > 1$$

$\therefore$ The pole $z = -2 + \sqrt{3}$ lies inside the contour C which is a simple pole.

Residue at $z = -2 + \sqrt{3}$ is

$$\lim_{z \rightarrow -2 + \sqrt{3}} (z + 2 - \sqrt{3}) f(z)$$

$$= \lim_{z \rightarrow -2 + \sqrt{3}} (z + 2 - \sqrt{3}) \cdot \frac{1}{z^2 + 4z + 1}$$

$$= \lim_{z \rightarrow -2 + \sqrt{3}} (z + 2 - \sqrt{3}) \frac{1}{(z + 2 - \sqrt{3})(z + 2 + \sqrt{3})}$$

$$= \lim_{z \rightarrow -2 + \sqrt{3}} \frac{1}{z + 2 + \sqrt{3}}$$

$$= \frac{1}{-2 + \sqrt{3} + 2 + \sqrt{3}}$$

$$= \frac{1}{2\sqrt{3}}$$

By Cauchy's residue theorem, we have from ①

$$\begin{aligned} \int_0^{2\pi} \frac{d\theta}{2 + \cos \theta} &= \frac{2}{i} \cdot 2\pi i (\text{Residue at } z = -2 + \sqrt{3}) \\ &= 4\pi \cdot \frac{1}{2\sqrt{3}} \\ &= \frac{2\pi}{\sqrt{3}} \quad \underline{\text{Ans.}} \end{aligned}$$

Problem-2: $\int_0^{2\pi} \frac{d\theta}{5 + 4\cos \theta}$

Hints: $\int_0^{2\pi} \frac{d\theta}{5 + 4\cos \theta} = \oint_C \frac{\frac{dz}{iz}}{5 + 4\frac{z^2+1}{z^2}} = \frac{1}{i} \oint_C \frac{dz}{5z^2 + 2z^2 + 2} = \frac{1}{i} \oint_C f(z) dz$

The poles of $f(z)$ are $z = -\frac{1}{2}, -2$

Residue at $z = -\frac{1}{2}$ is $\lim_{z \rightarrow -\frac{1}{2}} \left(z + \frac{1}{2} \right) \frac{1}{2z^2 + 5z + 2} = \frac{1}{3}$

By Cauchy's residue th^m, we get

$$\begin{aligned} \int_0^{2\pi} \frac{d\theta}{5 + 4\cos \theta} &= \frac{1}{i} \cdot 2\pi i (\text{Residue at } z = -\frac{1}{2}) \\ &= \frac{2\pi}{3} \end{aligned}$$

Problem-3: $\int_0^{2\pi} \frac{d\theta}{3+2\sin\theta}$

Hints: $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$
 $= \frac{z^2 - 1}{2iz}$

$$\int_0^{2\pi} \frac{d\theta}{3+2\sin\theta} = 2 \oint_C \frac{dz}{z^2 + 3iz - 1} = 2 \oint_C f(z) dz$$

$z = \frac{(-3+\sqrt{5})i}{2}, \frac{(-3-\sqrt{5})i}{2}$ are poles of $f(z)$.

Residue at $z = \frac{(-3+\sqrt{5})i}{2}$ is $\frac{1}{i\sqrt{5}}$

$\therefore$ By C-R-T, we get $\int_0^{2\pi} \frac{d\theta}{3+2\sin\theta} = 2 \cdot 2\pi i (\text{Residue at the pole})$
 $= 4\pi i \cdot \frac{1}{i\sqrt{5}}$
 $= \frac{4\pi}{\sqrt{5}}$

Problem-4: $\int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta} d\theta$

Hints: $z = e^{i\theta} \Rightarrow dz = ie^{i\theta} d\theta = iz d\theta$

$\Rightarrow d\theta = \frac{dz}{iz}; 0 \leq \theta \leq 2\pi$

$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$

$= \frac{1}{2} \left(z + \frac{1}{z} \right)$

$= \frac{z^2 + 1}{2z}$

$\therefore \int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta} d\theta$

$= \text{Real Part of } \int_0^{2\pi} \frac{\cos 2\theta + i \sin 2\theta}{5+4\cos\theta} d\theta$

$= \text{R.P of } \int_0^{2\pi} \frac{e^{i2\theta}}{5+4\cos\theta} d\theta$

$= \text{R.P of } \oint_C \frac{z^2}{5+4 \cdot \frac{z^2+1}{2z}} \cdot \frac{dz}{iz}$

$= \text{R.P of } \frac{1}{i} \oint_C \frac{z^2}{5z+2z^2+2} dz$

$= \text{R.P of } \frac{1}{i} \oint_C f(z) dz, (\text{say}) \quad \text{--- (1)}$

where $\oint_C f(z) dz = \frac{z^2}{2z^2+5z+2}$

The poles of $f(z)$ are $z = -\frac{1}{2}, -2$

Residue at $z = -\frac{1}{2}$ is

$$\lim_{z \rightarrow -\frac{1}{2}} (z + \frac{1}{2}) f(z) = \frac{1}{12}$$

$$\begin{aligned} \therefore \text{By G-R-T, we get } \int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta} d\theta &= \text{R.P. } \frac{1}{i} \cdot 2\pi i (\text{Residue at } z = -\frac{1}{2}) \\ &= \text{R.P. of } (2\pi \cdot \frac{1}{12}) \\ &= \frac{\pi}{6} \end{aligned}$$

H.W. Evaluate the following by using the method of contour integration.

i) $\int_0^{2\pi} \frac{d\theta}{3+2\cos\theta}$

ii) $\int_0^{2\pi} \frac{d\theta}{5+3\sin\theta}$

iii) $\int_0^{2\pi} \frac{d\theta}{2+\sin\theta}$

iv) $\int_0^{2\pi} \frac{d\theta}{(5-3\cos\theta)^2}$

v) $\int_0^{2\pi} \frac{d\theta}{(5-3\sin\theta)^2}$

$$(vi) \int_0^{2\pi} \frac{\cos 3\theta}{5-4\cos\theta} d\theta$$

$$(vii) \int_0^{2\pi} \frac{\sin 2\theta}{5-3\cos\theta} d\theta$$